



Code-Layout, Readability and Reusability

Date	02 July 2026
Team ID	XXXXXX
Project Name	Raising waters
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code	Yes	Proper indentation followed in Python and Flask files.
2	Proper File Structure	Files and folders are logically organized	Yes	Separate folders for templates, static files, model, dataset, and application.
3	Meaningful Variable Names	Variables reflect their purpose clearly	Yes	Variables such as rainfall, cloud_visibility, and prediction_result are descriptive.
4	Function / Method Names	Functions are descriptively named	Yes	Functions like predict_flood() and load_model() clearly indicate their purpose.

5	Code Comments	Inline and block comments explain logic	Yes	Important sections include comments for better understanding.
6	Modular Design	Code is split into reusable functions/modules	Yes	ML model loading, prediction, and preprocessing are implemented as separate modules.
7	No Redundant Code	Duplicate or unused code is removed	Yes	Unnecessary code has been removed to improve maintainability.
8	Error Handling	Exceptions and errors are handled gracefully	Yes	Basic input validation and exception handling are Implemented.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Prediction Model	Python, XGBoost	Flood prediction module	High
2	Data Preprocessing	Python, Pandas	Training and prediction	High
3	Flask Routes	Flask	User request handling	Medium
4	HTML Templates	HTML, CSS	User Interface	High
5	Model Loader	Pickle, Python	Application startup	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
--------	--------------	----------

Code Layout & Structure	5	Well-organized project with a clear folder structure.
Readability	5	Meaningful names, proper formatting, and comments improve readability.
Reusability	5	Modules can be reused for future enhancements and deployment.
Documentation / Comments	5	Code is documented with essential comments; more detailed documentation can be added.
Overall Score	5 / 5	The project follows good coding practices with a modular and maintainable design.